NPTEL Online Certification Courses

Indian Institute of Technology Kharagpur

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Introduction to Internet of Things Assignment-

Week 9

TYPE OF QUESTION: MCQ/MSQ

Number of questions: **15** **Total marks: 15 X 1= 15**

QUESTION 1:

Which of the following can be used to install OpenStack?

a. devstack
b. stack
c. Swift
d. All of these

Correct Answer: a. devstack

Detailed Solution: devstack can be used to install OpenStack. (Please refer Lecture 41@8:01)

QUESTION 2:

Which of the following is/are the advantages of cloud computing?

a. Elasticity
b. Pay-per-use
c. Self Service
d. All of the above

Correct Answer: d. All of the above

Detailed Solution: The advantages of cloud computing include Elasticity, Pay-per-use and Self-

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Service. (Please refer Lecture 42@6:45)

QUESTION 3:

Select the statement(s) that denote the limitations of WSNs.

Statement I: Procurement

Statement II: Deployment

Statement III: Maintenance

a. Statement I

b. Statement II

c. Statements I, II, and III

d. None of these

Correct Answer: c. Statements I, II, and III

Detailed Solution: The limitations of WSNs are Procurement, Deployment and Maintenance. (Please refer Lecture 42@10:52)

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QUESTION 4:

Which of the following plays a role in the sensor-cloud architecture?

a. End-users

b. Sensor-owner

c. Sensor-Cloud Service Provider

d. All of these

Correct Answer: d. All of these

Detailed Solution: End users, sensor-owner, sensor-cloud service provider are the actors in sensor-cloud. (Please refer Lecture 42@14:18)

QUESTION 5:

The managerial role is played by _____ in sensor-cloud architecture.

a. End-users

b. Sensor-Cloud Service Provider

c. Neither a nor b

d. Both a and b

Correct Answer: b. Sensor-Cloud Service Provider

Detailed Solution: Sensor-Cloud Service Provider plays the managerial role in

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sensor-cloud architecture. (Please refer Lecture 42@14:29)

QUESTION 6:

Which of the following is/are the components of the work-flow of the sensor-cloud.

a. Virtual Sensor Manager

b. Virtual Sensor Controller

c. Resource Manager

d. All of these

Correct Answer: d. All of these

Detailed Solution: Virtual Sensor Manager, Virtual Sensor Controller and Resource Manager are all of the components of the work-flow of the sensor-cloud. (Please refer Lecture 42@17:02)

QUESTION 7:

Who coined the term Fog computing?

a. IBM

b. CISCO

c. Both a and b

d. Neither a nor b

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Correct Answer: b. CISCO

Detailed Solution: CISCO coined the term Fog computing. (Please refer Lecture 44@4:18)

QUESTION 8:

What of the following is responsible for allocation and deallocation of the resources?

a. Sensor manager

b. Sensor controller

c. Sensor resource pool

d. None of these

Correct Answer: c. Sensor resource pool

Detailed Solution: Sensor resource pool is responsible for allocation and deallocation of the resources (Please refer Lecture 42@16:11)

QUESTION 9:

The sensor owner can only deploy homogeneous physical sensor nodes in the sensor cloud.

a. True

b. False

Correct Answer: b. False

Detailed Solution: The sensor owner can deploy homogeneous and heterogeneous physical sensor nodes in the sensor cloud. (Please refer Lecture 42@15:03)

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QUESTION 10:

What is the function of Internal Cache?

a. Handles requests from end-user
b. Takes decision whether the data should be provided directly to the end user
c. Neither a nor b
d. Both a and b

Correct Answer: d. Both a and b

Detailed Solution: Internal Cache handles requests from end-user and also takes decision whether the data should be provided directly to the end user (Please refer Lecture 43@21:06)

QUESTION 11:

What is the other term for fog computing?

a. Foglog

b. Cloud

c. Dew

d. Fogging

Correct Answer: d. Fogging

Detailed Solution: The other term for fog computing is Fogging. (Please refer Lecture 44@4:19)

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QUESTION 12:

Fill in the blanks. Fog computing is an intermediate layer between _____ and _____.

a. Dew and devices

b. Cloud and devices

c. Cloud and server

d. None of these

Correct Answer: b. Cloud and devices

Detailed Solution: Fog computing is an intermediate layer between Cloud and devices. (Please refer Lecture 44@6:40)

QUESTION 13:

Bandwidth refers to the bit-rate of data during analysis.

a. True

b. False

Correct Answer: b. False

Detailed Solution: Bandwidth refers to the bit-rate of data during transmission. (Please refer Lecture 44@17:03)

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QUESTION 14:

Which of the following is used by the fog applications to increase the speed of service accessibility?

a. Low power

b. Good network connection

c. Accelerators

d. Analytics

Correct Answer: c. Accelerators

Detailed Solution: Accelerators increase the speed of service accessibility. (Please refer to Page 260, Chapter 11, Introduction to IoT. S. Misra, A. Mukherjee, and A. Roy, 2020. Cambridge University Press.)

QUESTION 15:

Fill in the blank. The concept of _____ enables physical hardware to be shared among multiple entities.

a. Hardware virtualization

b. Software virtualization

c. Module virtualization

d. All of these

Correct Answer: a. Hardware virtualization

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Detailed Solution: The concept of hardware virtualization enables physical hardware to be shared among multiple entities. (Please refer to Page 262, Chapter 11, Introduction to IoT. S. Misra, A. Mukherjee, and A. Roy, 2020. Cambridge University Press.)

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